

Department of Electrical and Information Engineering
EE7218/EC7207 High Performance Computing
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Project title:	Architectural Scaling of Iterative Linear Solvers: A Comparative Analysis of Parallel Jacobi and Red-Black Gauss-Seidel Implementations
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Project Description :	
1. Problem Statement and Scope The solution of large-scale systems of linear equations ($Ax = b$) is a fundamental requirement in fields such as structural engineering, climate modeling, and Markov chain analysis. While direct solvers like Gaussian Elimination are robust, iterative solvers specifically the Jacobi and Gauss-Seidel (GS) methods are often preferred for massive, sparse systems due to their lower memory footprint. This project will implement and optimize these solvers across a four-tier parallel hierarchy to evaluate their performance and numerical stability. 2. Algorithmic Approach The project will focus on solving a strictly diagonally dominant system, ensuring convergence for both methods. • Jacobi Method: This method updates each component $x_i^{(k+1)}$ using only values from the previous iteration x^k, making it naturally parallelizable. • Gauss-Seidel Method: To address the inherently sequential nature of GS (which typically uses updated values immediately), we will implement Red-Black (Checkerboard) Ordering. This technique reorders the unknowns into two independent sets ("red" and "black") to allow for parallel updates while maintaining faster convergence than Jacobi.	